

Polifenoles de calafate

Berberis microphylla G. Forst

Estructuras químicas para COSMO-RS

40 estructuras discretas

Antocianina — 10

Flavonol — 11

Flavona — 2

Flavan-3-ol — 2

Ác. Hidroxicinámico — 8

Ác. Hidroxibenzoico — 4

Tanino Condensado — 2

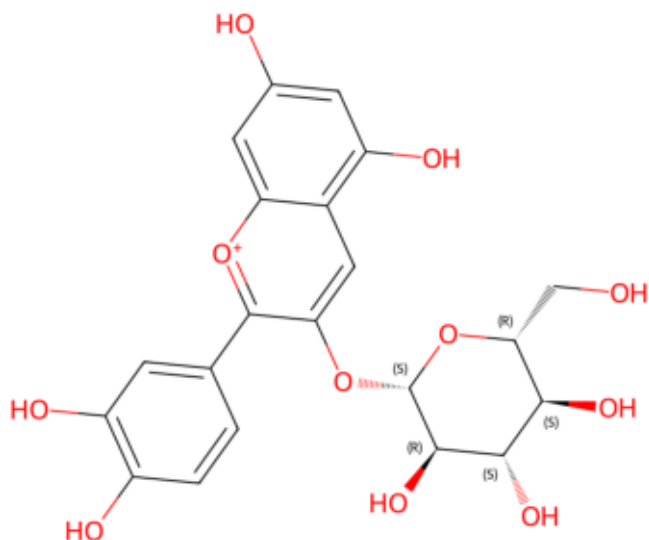
Alcaloide isoquinolínico — 1

Las antocianinas se representan como CATIÓN FLAVILIO (carga +1), que es la especie dominante a pH 2-4 (rango de los NADES de este trabajo).

Fórmula, peso molecular y carga verificados computacionalmente (RDKit).
La estereoquímica NO está verificada: contrastar por InChIKey contra PubChem antes del paso QM.

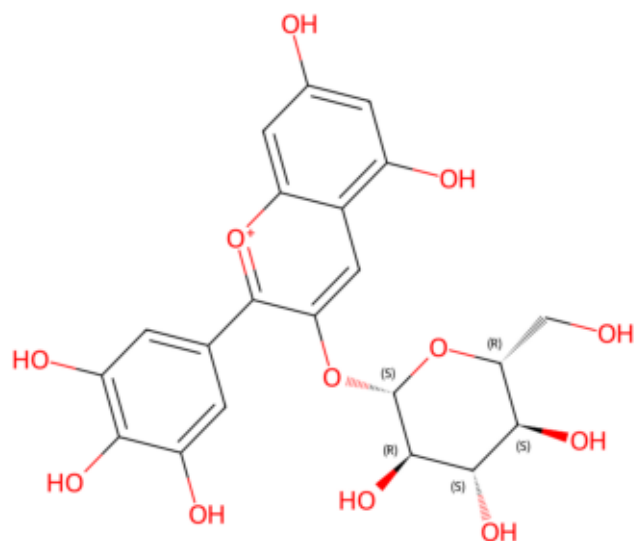
Fuente de la lista: Ruiz et al. (2024) Horticulturae 10, 458 (fruto);
Mocan et al. (2017) y Muñoz et al. (2011) para hojas y tallo,
con asignaciones generalizadas de Berberis spp.

Cianidín-3-glucósido



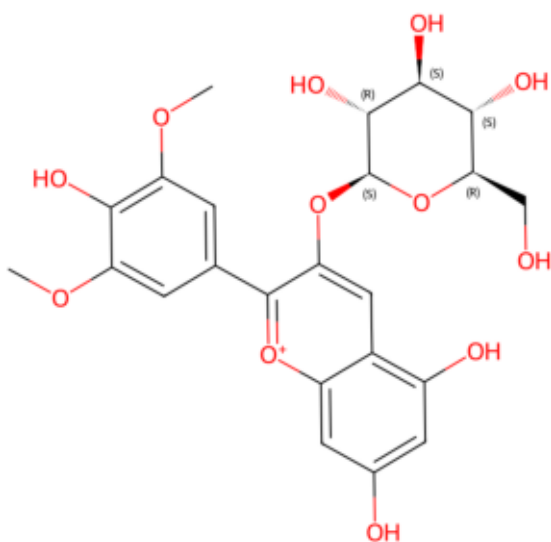
Antocianina · C₂₁H₂₁O₁₁⁺ · 449.39 g/mol · carga +1
fruto · BKWHW5ONKIEUFF-GOHP9BCVSA-O

Delphinidín-3-glucósido



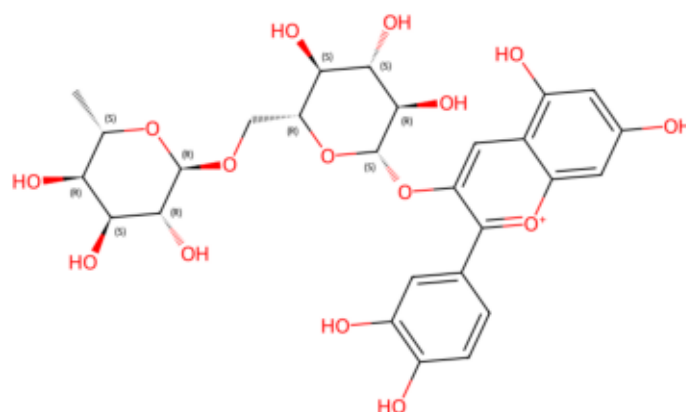
Antocianina · C₂₁H₂₁O₁₂⁺ · 465.39 g/mol · carga +1
fruto · XFNHR2QLDPAYIL-PEVILNPASA-O

Malvidín-3-glucósido



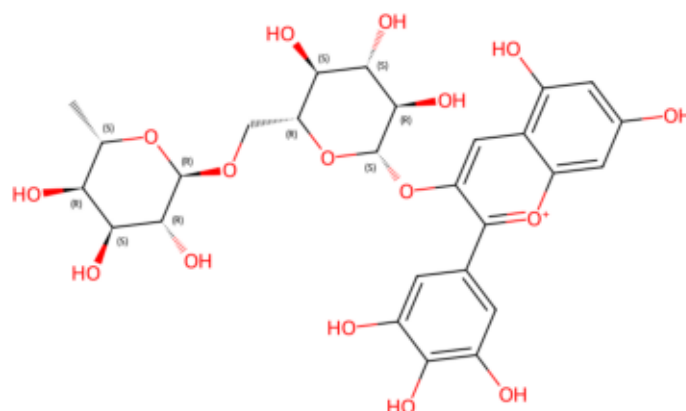
Antocianina · C₂₃H₂₅O₁₂⁺ · 493.44 g/mol · carga +1
fruto · PXUQTDZNOHRWLI-OXUVVOBNSA-O

Cianidín-3-rutinosido



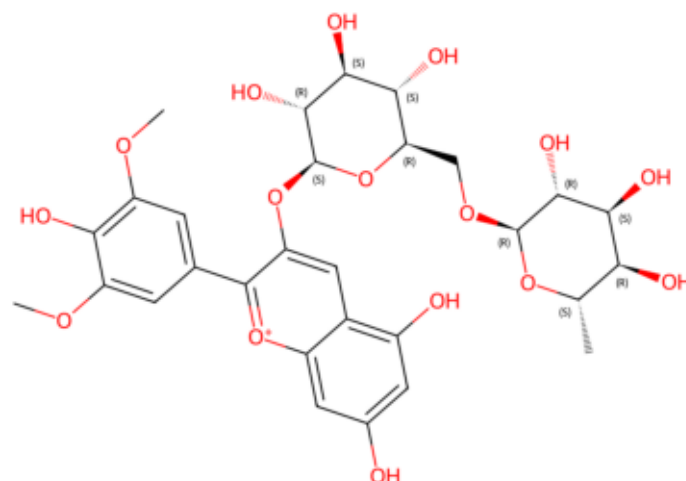
Antocianina · C₂₇H₃₁O₁₅⁺ · 595.53 g/mol · carga +1
fruto · U5NPILRDBDVIAO-ACXVBFAASA-O

Delphinidín-3-rutinosido



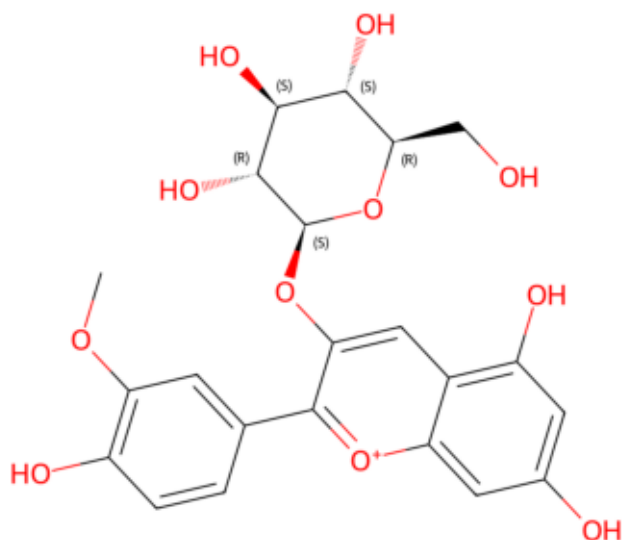
Antocianina · C₂₇H₃₁O₁₆⁺ · 611.53 g/mol · carga +1
fruto · PLKUTZNSKRWCCA-OXVORGPNASA-O

Malvidín-3-rutinosido



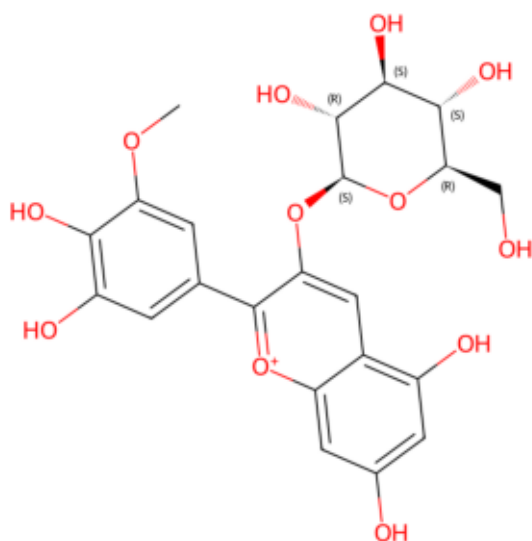
Antocianina · C₂₉H₃₅O₁₆⁺ · 639.58 g/mol · carga +1
fruto · YCDMGUCUGMVVWAB-YAGJAPKFSA-O

Peonidín-3-glucósido



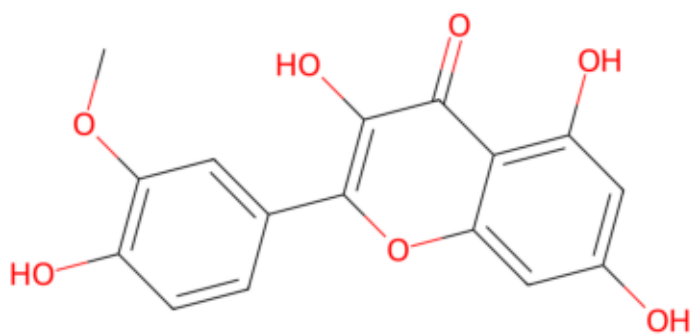
Antocianina · C₂₂H₂₃O₁₁⁺ · 463.42 g/mol · carga +1
fruto · ZZWPMFROUJHAKY-OUHKCGNVSA-O

Petunidín-3-glucósido



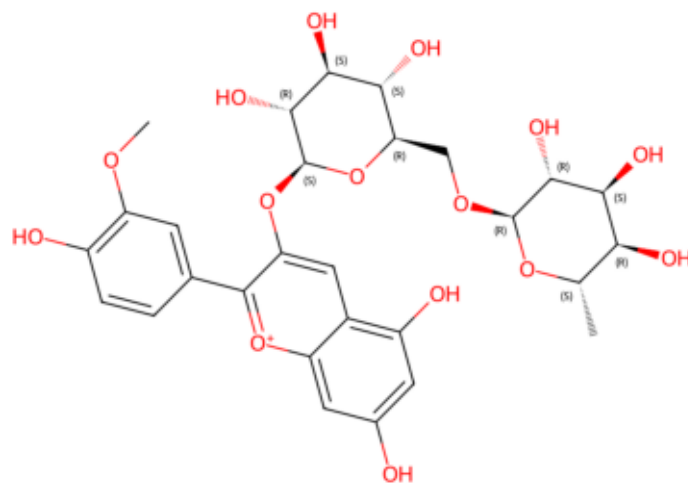
Antocianina · C₂₂H₂₃O₁₂⁺ · 479.41 g/mol · carga +1
fruto · CCQDWIRWKWIUKK-QKYRYQKWSA-O

Isorhamnetina



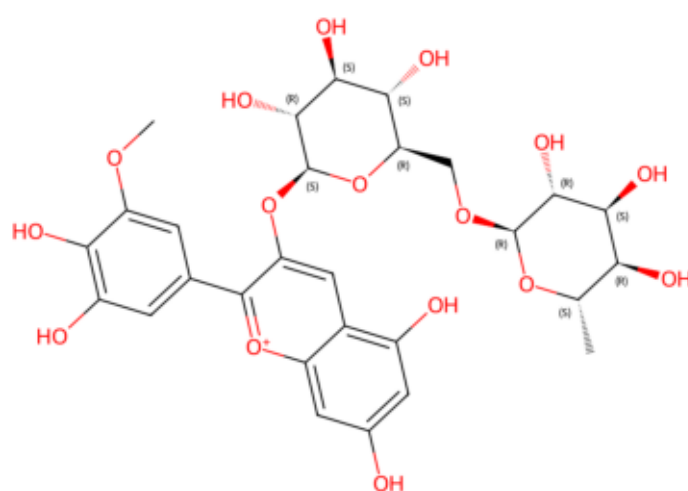
Flavonol · C₁₆H₁₂O₇ · 316.27 g/mol
hojas · IZQSVBPOUDKVDZ-UHFFFAOYSA-N

Peonidín-3-rutinosido



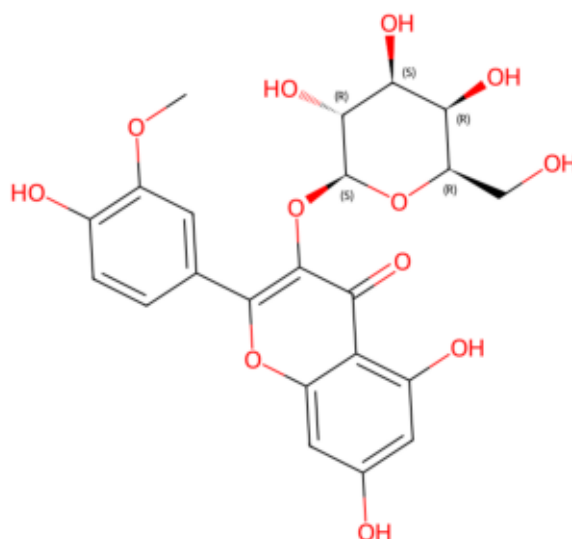
Antocianina · C₂₈H₃₃O₁₅⁺ · 609.56 g/mol · carga +1
fruto · ONOVTPEFYSRBU-LVOFENWISA-O

Petunidín-3-rutinosido



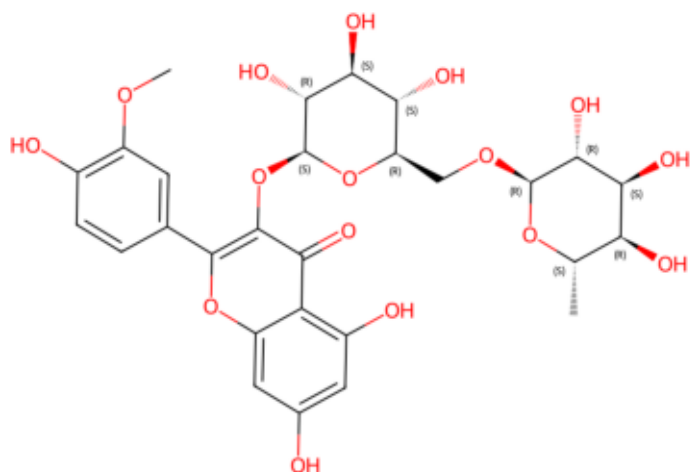
Antocianina · C₂₈H₃₃O₁₆⁺ · 625.56 g/mol · carga +1
fruto · IWTWNOZWPPSQB-GBDNGMYS-A-O

Isorhamnetin-3-galactosido



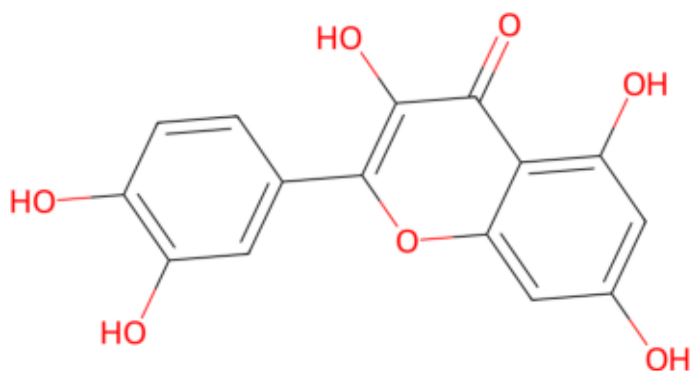
Flavonol · C₂₂H₂₂O₁₂ · 478.41 g/mol
fruto · CQLRUUIRRZYHHS-UVHBULKNSA-N

Isorhamnetín-3-rutinosido



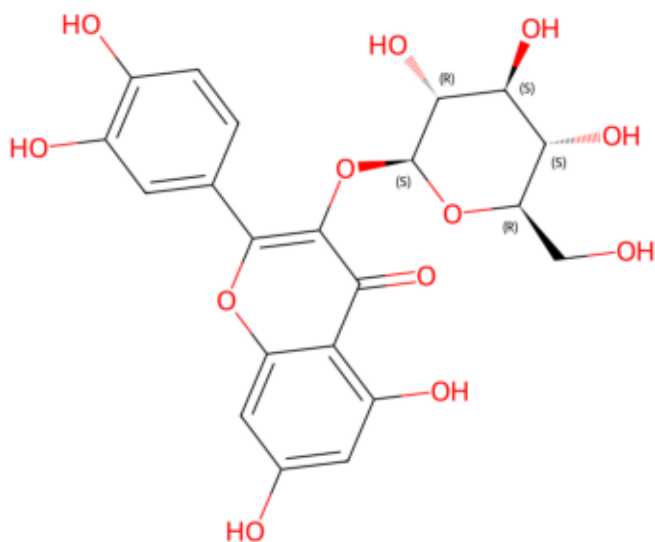
Flavonol · C₂₈H₃₂O₁₆ · 624.55 g/mol
fruto · UIDGXJNOLKLM-WDJQYGGQPSA-N

Quercetina



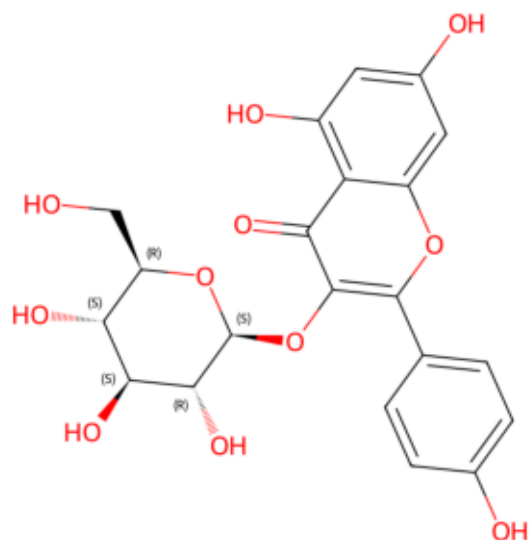
Flavonol · C₁₅H₁₀O₇ · 302.24 g/mol
hojas · REFIWTPEDVJILY-UHFFFAOYSA-N

Quercetín-3-glucósido



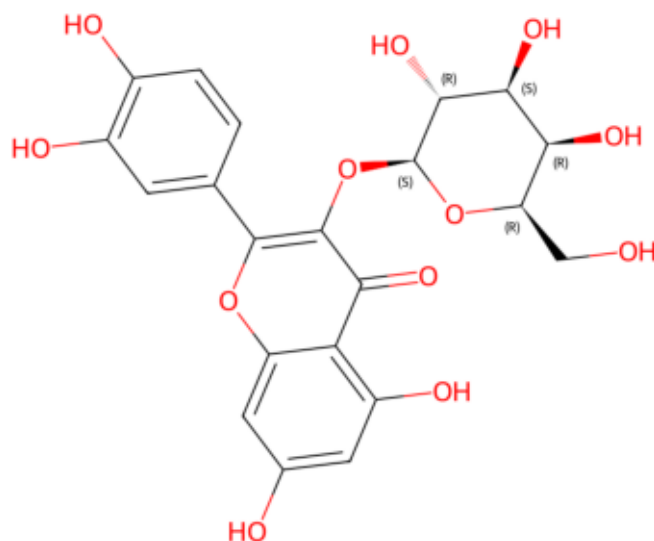
Flavonol · C₂₁H₂₀O₁₂ · 464.38 g/mol
fruto · OVSQVDMCBVZWGM-QSOFNFLRSA-N

Kaempferol-3-glucósido



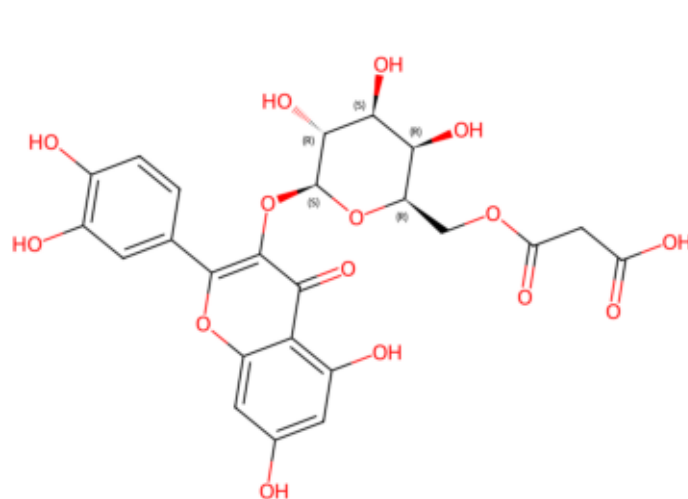
Flavonol · C₂₁H₂₀O₁₁ · 448.38 g/mol
hojas · IPIUKWEQWCBDDOP-QSOFNFLRSA-N

Quercetín-3-galactósido



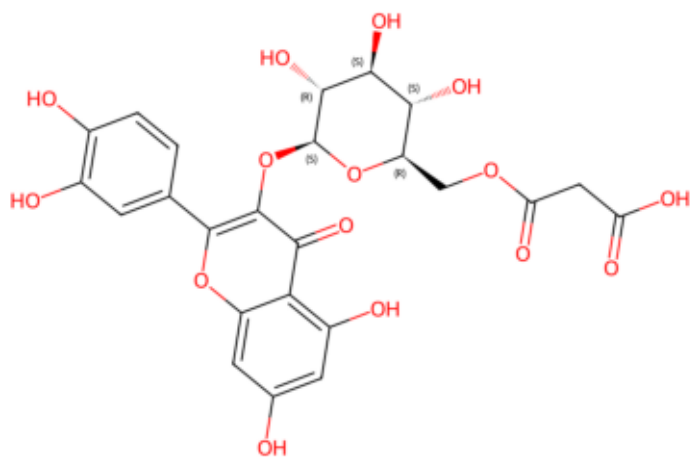
Flavonol · C₂₁H₂₀O₁₂ · 464.38 g/mol
fruto · OVSQVDMCBVZWGM-DTGCBRNFSAN

Quercetín-3-malonilgalactósido



Flavonol · C₂₄H₂₂O₁₅ · 550.43 g/mol
fruto · NBQPHANHNTWDM-LYYZAYPPSA-N

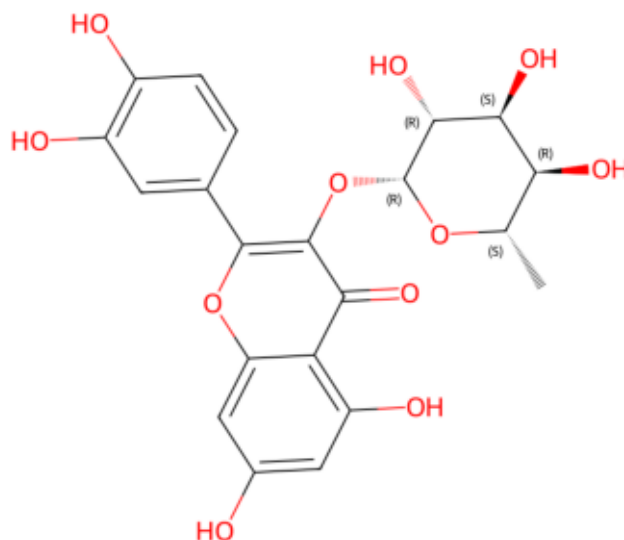
Quercetín-3-malonilglucósido



Flavonol · C₂₄H₂₂O₁₅ · 550.43 g/mol
fruto · NBOPHANHTWDM-LIKBSQBPSA-N

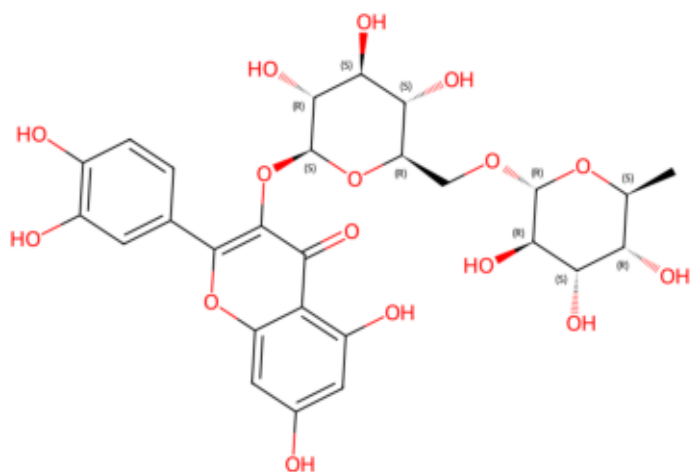
Quercetín-3-rutinosido (Rutina)

Quercetín-3-rhamnosido



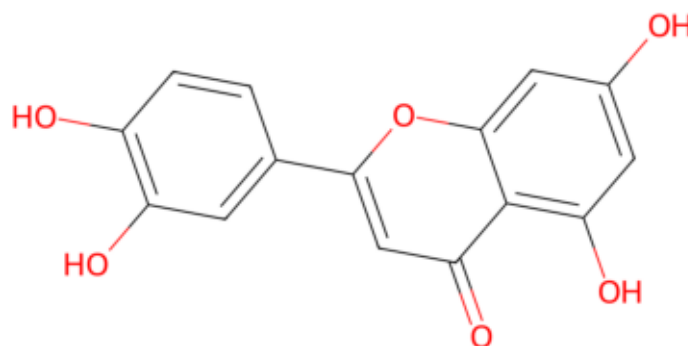
Flavonol · C₂₁H₂₀O₁₁ · 448.38 g/mol
fruto · OXGUCJVFQIWWQIYHQAYEMVSA-N

Luteolina



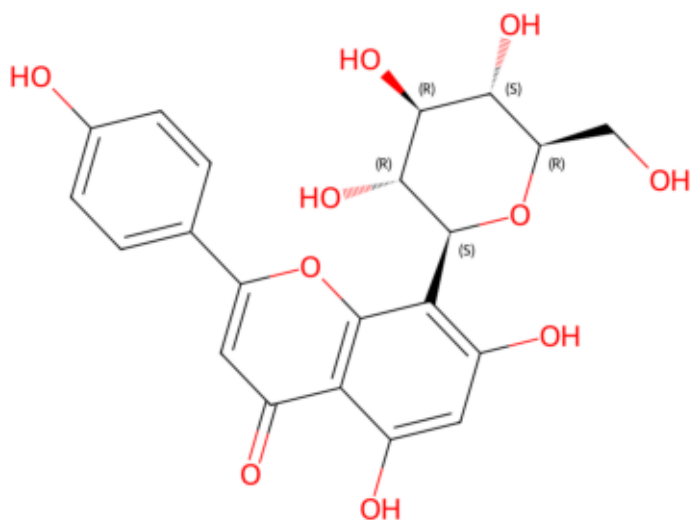
Flavonol · C₂₇H₃₀O₁₆ · 610.52 g/mol
fruto|hojas · IKGYIPEFEMILURG-FYIKAPSBASA-N

Vitexina

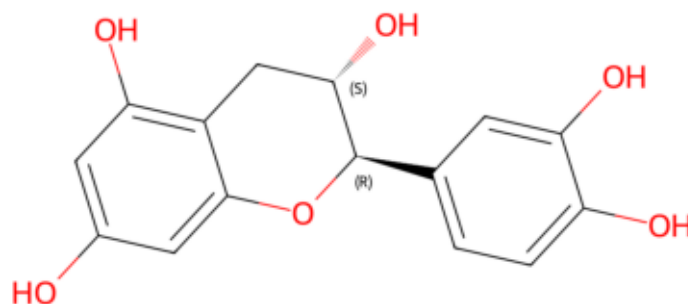


Flavona · C₁₅H₁₀O₆ · 286.24 g/mol
hojas · IQPNAANSBPBGFO-UHFFFAOYSA-N

Catequina

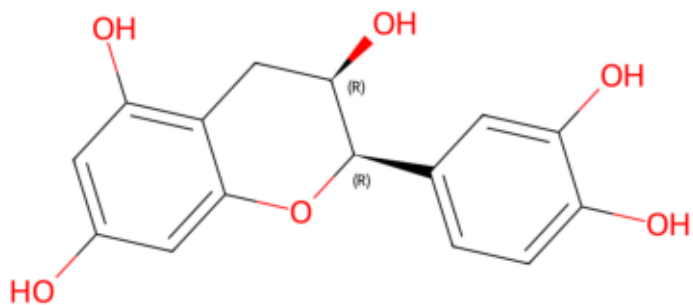


Flavona · C₂₁H₂₀O₁₀ · 432.38 g/mol
hojas · SGEWCQFRYRRZDC-VPRICQMDSA-N



Flavan-3-ol · C₁₅H₁₄O₆ · 290.27 g/mol
hojas|tallo · PFTAWBLQPZVEMU-DZGCQCFKSA-N

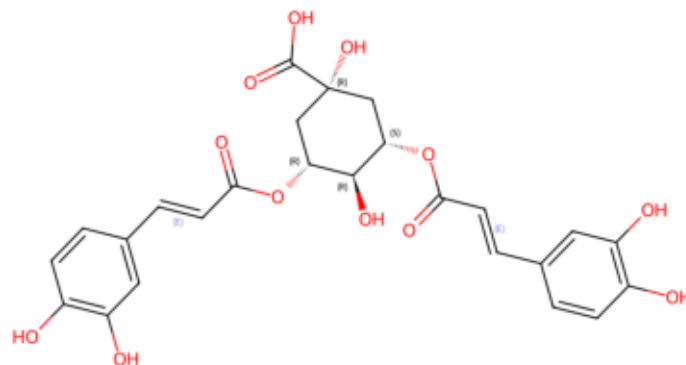
Epicatequina



Flavan-3-ol · C₁₅H₁₄O₆ · 290.27 g/mol
hojas|tallo · PFTAWBLOPZYEMU-UKBRQHHQSA-N

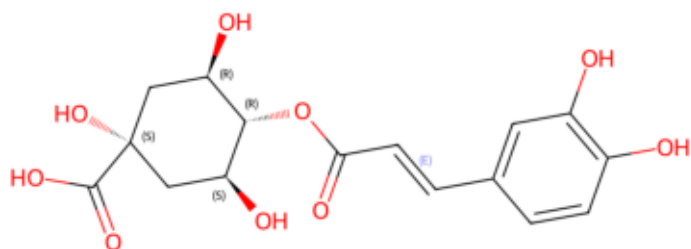
3-Cafeoilquínico

3,5-Dicafeoilquínico



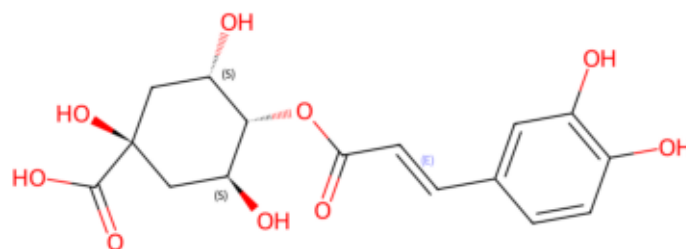
Ác. Hidroxicinámico · C₂₅H₂₄O₁₂ · 516.46 g/mol
fruto · KPZPCUWA/BGOTN7-CPYHIKAUSA-N

4-Cafeoilquínico



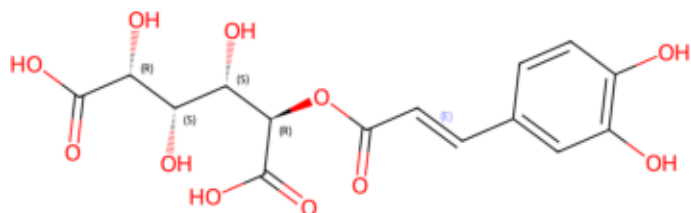
Ác. Hidroxicinámico · C₁₆H₁₈O₉ · 354.31 g/mol
fruto · GYEFKZTYAFCIR-BHLOTTMOSA-N

Cafeoilglucarico (isómeros A-D)

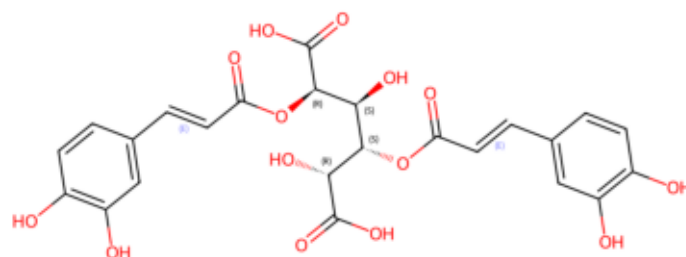


Ác. Hidroxicinámico · C₁₆H₁₈O₉ · 354.31 g/mol
fruto · GYEFKZTYAFCIR-ZZYVFCPTSA-N

Dicafeoilglucarico

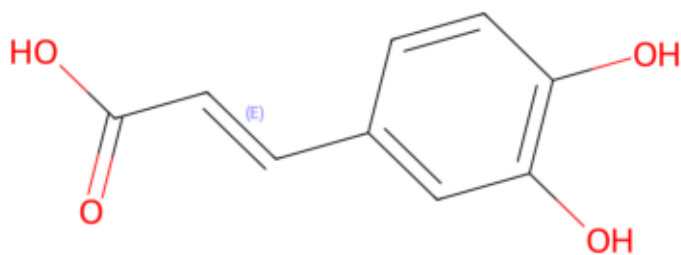


Ác. Hidroxicinámico · C₁₅H₁₆O₁₁ · 372.28 g/mol
fruto · OMIXKOJEIRRAJW-HLWOJZJGSA-N



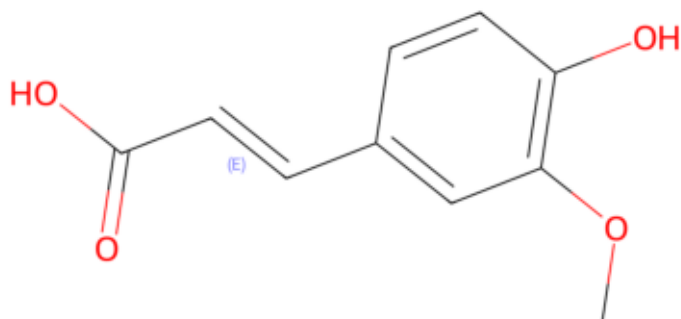
Ác. Hidroxicinámico · C₂₄H₂₂O₁₄ · 534.43 g/mol
fruto · ZQGMKFHEDFDRLZ-BYCAZERCSA-N

Ácido Cafeico



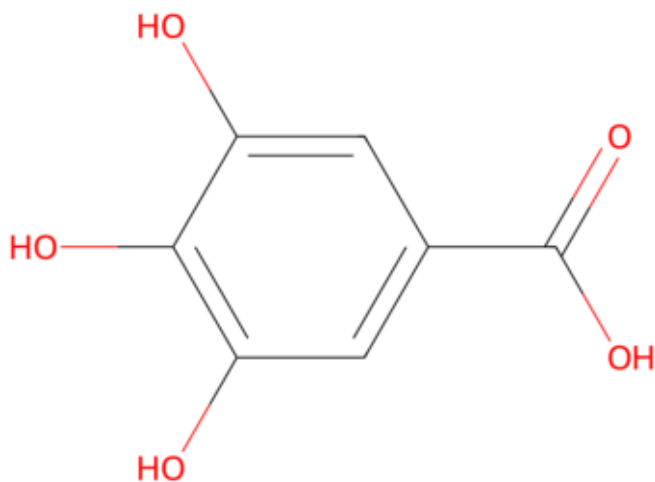
Ác. Hidroxicinámico · C₉H₈O₄ · 180.16 g/mol
hojas|tallo · CAUPRYCQNGVQAS-DUXPYHPUSA-N

Ácido Ferúlico



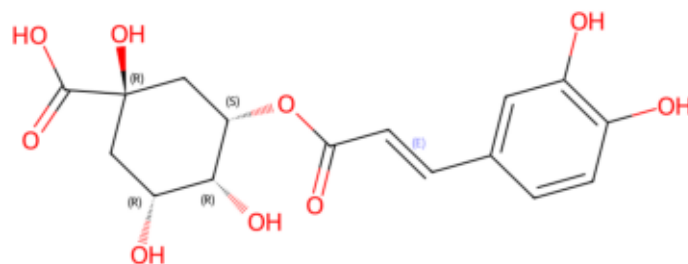
Ác. Hidroxicinámico · C₁₀H₁₀O₄ · 194.19 g/mol
hojas · KSEBNYQBYZTQHS-HWKANZROSA-N

Ácido Galico



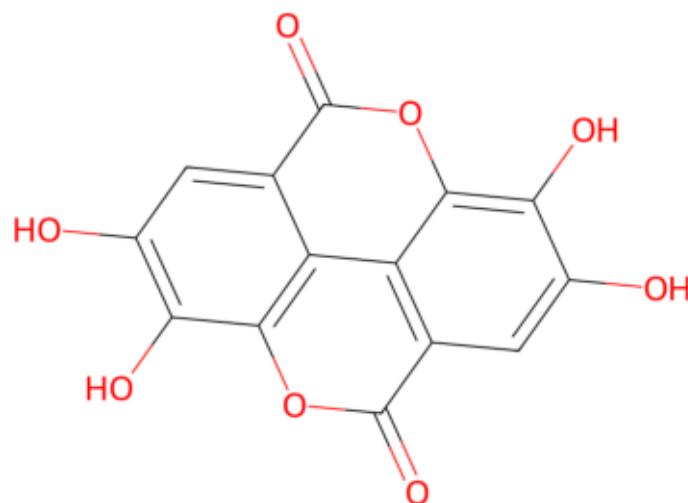
Ác. Hidroxibenzoico · C₇H₆O₅ · 170.12 g/mol
hojas|tallo · LNTHITQWFMADLM-UHFFFAOYSA-N

Ácido Clorogénico (5-cafeoilquínico)



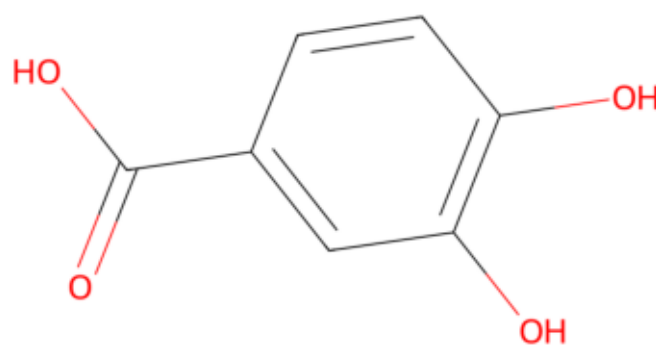
Ác. Hidroxicinámico · C₁₆H₁₈O₉ · 354.31 g/mol
fruto|hojas|tallo · CQWBNITMFTYNAD-QEPAVKMUSA-N

Ácido Elágico



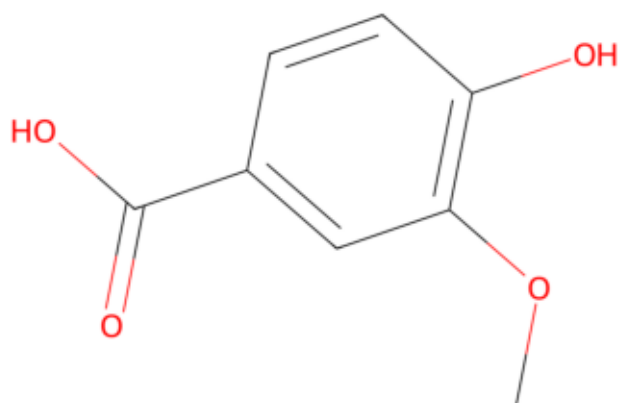
Ác. Hidroxibenzoico · C₁₄H₆O₈ · 302.19 g/mol
fruto|tallo · AESBNELWKVMVRR-LHFFFAOYSA-N

Ácido Protocatequico



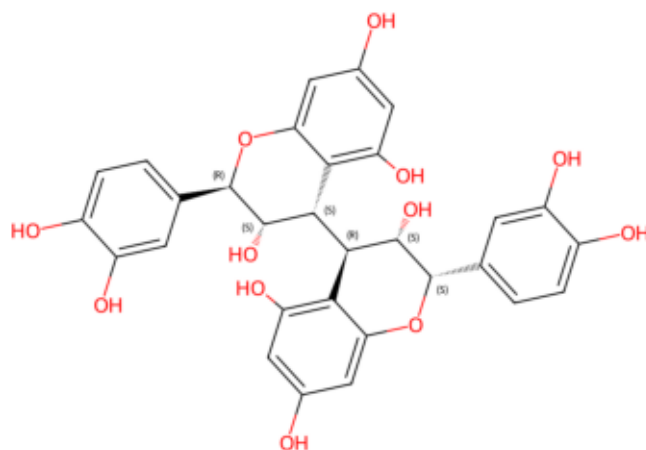
Ác. Hidroxibenzoico · C₇H₆O₄ · 154.12 g/mol
tallo · YQUVCSBJEUQKSH-UHFFFAOYSA-N

Ácido Vanílico



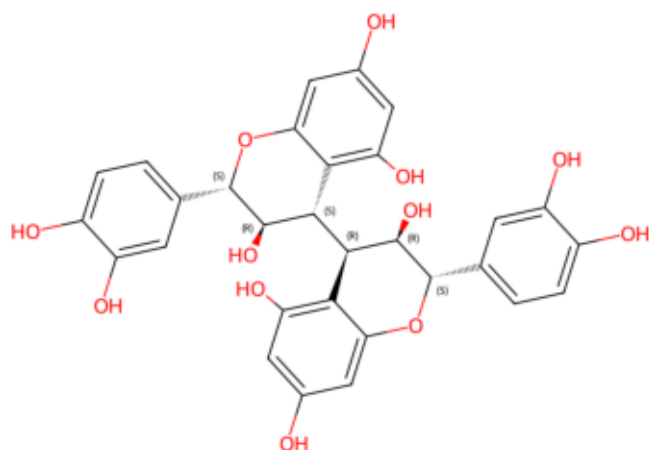
Ác. Hidroxibenzoico · C₈H₈O₄ · 168.15 g/mol
tallo · WKOLLVMJNQIZCI-UHFFFAOYSA-N

Procianidina B1



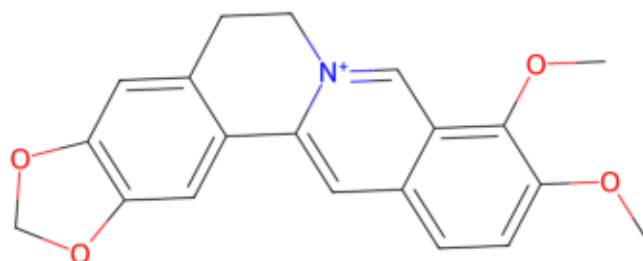
Tanino Condensado · C₃₀H₂₆O₁₂ · 578.53 g/mol
fruto|tallo · UPSCCKSYXWBZPX-FSIJYMTBSA-N

Procianidina B2



Tanino Condensado · C₃₀H₂₆O₁₂ · 578.53 g/mol
fruto|hojas|tallo · UPSCCKSYXWBZPX-KWKLTRBOSA-N

Berberina (*alcaloide, no polifenol)



Alcaloide isoquinolínico · C₂₀H₁₈NO₄⁺ · 336.37 g/mol · carga +1
tallo · YBHILYKTIRIUTE-UHFFFAOYSA-N